

12. Consider the following statements in respect of non-singular matrix A of order n :
1. $A(\text{adj } A^T) = A(\text{adj } A)^T$
 2. If $A^2 = A$, then A is identity matrix of order n
 3. If $A^3 = A$, then A is identity matrix of order n
- Which of the statements given above are correct?
- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
13. How many four-digit natural numbers are there such that all of the digits are even?
- (a) 625 (b) 500 (c) 400 (d) 256
14. If $\omega \neq 1$ is a cube root of unity, then what are the solutions of $(z - 100)^3 + 1000 = 0$?
- (a) $10(1 - \omega)$, $10(10 - \omega^2)$, 100
(b) $10(10 - \omega)$, $10(10 - \omega^2)$, 90
(c) $10(1 - \omega)$, $10(10 - \omega^2)$, 1000
(d) $(1 + \omega)$, $(10 + \omega^2)$, -1
15. What is $(1 + i)^4 + (1 - i)^4$ equal to, where $i = \sqrt{-1}$?
- (a) 4 (b) 0 (c) -4 (d) -8
16. Consider the following statements in respect of a skew-symmetric matrix A of order 3:
1. All diagonal elements are zero.
 2. The sum of all the diagonal elements of the matrix is zero.
 3. A is orthogonal matrix.
- Which of the statements given above are correct?
- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
17. Four digit numbers are formed by using the digits 1, 2, 3, 5 without repetition of digits. How many of them are divisible by 4?
- (a) 120 (b) 24 (c) 12 (d) 6
18. What is the remainder when 2^{120} is divided by 7?
- (a) 1 (b) 3 (c) 5 (d) 6
19. For what value of n is the determinant
- $$\begin{vmatrix} C(9,4) & C(9,3) & C(10,n-2) \\ C(11,6) & C(11,5) & C(12,n) \\ C(m,7) & C(m,6) & C(m+1,n+1) \end{vmatrix} = 0$$
- for every $m > n$?
- (a) 4 (b) 5 (c) 6 (d) 7
20. If ABC is a triangle, then what is the value of the determinant
- $$\begin{vmatrix} \cos C & \sin B & 0 \\ \tan A & 0 & \sin B \\ 0 & \tan(B+C) & \cos C \end{vmatrix} = ?$$
- (a) -1 (b) 0 (c) 1 (d) 3
21. What is the number of different matrices, each having 4 entries that can be formed using 1, 2, 3, 4 (repetition is allowed)?
- (a) 72 (b) 216 (c) 254 (d) 768
22. Let $A = \{x \in R: -1 < x < 1\}$. Which of the following is/are bijective functions from A to itself?
1. $f(x) = x|x|$
 2. $g(x) = \cos(\pi x)$
- Select the correct answer using the code given below:
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
23. Let R be a relation on the open interval $(-1, 1)$ and is given by $R = \{(x, y): |x + y| < 2\}$. Then which one of the following is correct?
- (a) R is reflexive but neither symmetric nor transitive
(b) R is reflexive and symmetric but not transitive
(c) R is reflexive and transitive but not symmetric
(d) R is an equivalence relation
24. For any three non-empty sets A, B, C , what is $(A \cup B) - \{(A - B) \cup (B - A) \cup (A \cap B)\}$ equal to?
- (a) Null set (b) A
(c) B (d) $(A \cup B) - (A \cap B)$
25. If a, b, c are the sides of triangle ABC , then what is
- $$\begin{vmatrix} a^2 & b \sin A & c \sin A \\ b \sin A & 1 & \cos A \\ c \sin A & \cos A & 1 \end{vmatrix}$$
- equal to?
- (a) Zero
(b) Area of triangle
(c) Perimeter of triangle
(d) $a^2 + b^2 + c^2$
26. If a, b, c are in AP; b, c, d are in GP; c, d, e are in HP, then which of the following is/are correct?
1. a, c and e are in GP
 2. $\frac{1}{a}, \frac{1}{c}, \frac{1}{e}$ are in GP
- Select the correct answer using the code given below:
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
27. What is the number of solutions of $\log_4(x - 1) = \log_2(x - 3)$?
- (a) Zero (b) One (c) Two (d) Three